

MirroringWay: Designing an Inter-World Transition Method with Materialized Mirror Imagery

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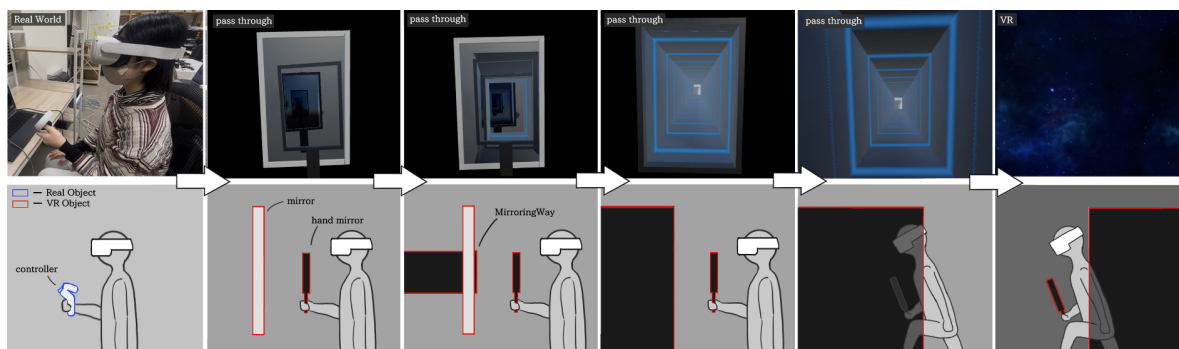


Figure 1: Overview of the MirroringWay experience flow.

ABSTRACT

This study explores a novel transition method for enabling experiences that traverse multiple types of worlds, including real, virtual reality (VR), and augmented reality (AR). As such an approach, we designed MirroringWay, a transition method that employs a mirror as the mediating element. MirroringWay reproduces within VR mirror-generated imagery resembling an endlessly extending space produced by a facing-mirror configuration and materializes it as a traversable transition space. In this paper, we conducted a demo-oriented user study focusing on experiences that involve transitions across multiple heterogeneous spaces. The results suggest that MirroringWay enables spatial transitions to be understood as a reasoned movement experience and has the potential to endow the transition itself with meaning and enjoyment.

Index Terms: VR, mirror, perception and cognition.

1 INTRODUCTION

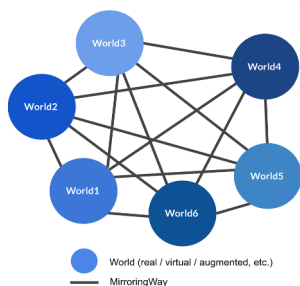


Figure 2: Conceptual structure of inter-world transitions enabled by MirroringWay, illustrating topological interconnections across worlds.

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With recent advances in XR technologies, users can easily move between different spaces—such as real, virtual reality (VR), and augmented reality (AR)—through passthrough displays and by putting on or removing head-mounted displays (HMDs). However, during such transitions, visual discontinuities often arise, reducing the sense of immersion. Previous studies have investigated visual transitions between VR and AR environments and highlighted the importance of continuity in user experience [2]. Nevertheless, the design of transition methods that explicitly assume traversal across multiple different worlds remains an open challenge. Focusing on continuity, this study addresses how transitions across different worlds can be designed as perceptual processes rather than scene switches. As one approach to this challenge, we propose MirroringWay, a transition method that employs a mirror as the mediating element. Mirrors are familiar objects in the real world; however, prior research has shown that they can evoke non-ordinary perceptual experiences, such as altered perceptions of one's own body through changes in self-representation [1]. Based on these characteristics, mirrors function as interfaces that connect the real world with different types of spaces. In this paper, as a fundamental evaluation of MirroringWay, we investigate whether mirror-based movement can function as a method for spatial transitions.

2 DESIGN CONCEPT

Rather than interpreting the infinite mirror illusion as an extension of Euclidean depth, MirroringWay reframes it as a passage created through spatial recursion, where repeated mirror imagery functions as a topological connection between distinct worlds.

In XR environments, spatial transitions are often presented as button-based operations or abrupt scene switches. However, such approaches do not always allow users to naturally perceive the process of movement itself. In this study, we consider the continuous presentation of user actions and spatial changes to be essential for a convincing transition experience. Based on this perspective, we designed MirroringWay, a transition method that employs a mirror as the mediating element and uses the act of passing through a mirror as the starting point of the transition. MirroringWay constructs a transition experience in which users can perceive that they have moved into another world through the act of passing through a mirror. The sequential transformation of mirror imagery into an

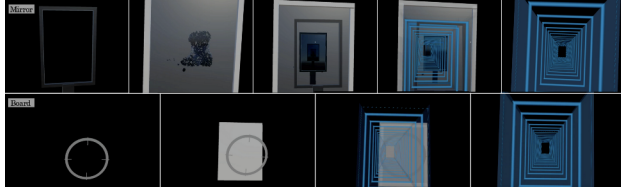


Figure 3: Materializing the Transition Space in the Board and Mirror Conditions.

enterable space functions as the perceptual trigger of the transition. MirroringWay supports transitions across multiple worlds, including real, virtual, and augmented environments (Figure 2).

3 EXPERIENCE DESIGN AND IMPLEMENTATION

This section describes the experience design and implementation of the proposed method, MirroringWay. Participants wear an HMD and initially view the passthrough environment. During the experience, a virtual hand mirror attached to the participant's right hand follows their movement, while a full-body virtual mirror is placed in the environment. The surface of the virtual mirror displays the participant's appearance captured by a depth camera, presented as the participant's avatar within VR. When the participant orients the hand mirror toward the full-body mirror, a facing-mirror state is established, producing a visual effect in which the reflected imagery appears to extend into depth. While in this state, pressing the controller's trigger button generates a transition space within the reflected imagery, creating the visual impression that the mirror imagery has materialized. We define this materialized transition space as MirroringWay. Immediately after generation, MirroringWay appears at a scale similar to the reflected imagery and gradually expands to align with the full-body mirror, allowing the participant to pass through it. By moving forward through MirroringWay, the participant transitions into another world.

4 DEMO DESIGN AND OBSERVATIONS

This study conducted a demo-oriented user study to examine how MirroringWay influences experiences involving transitions across multiple spaces. Twelve participants of diverse ages and genders took part in the study, comparing three transition methods and evaluating enjoyment, perceived continuity of spatial transitions, immersion in the destination space, operability, and motion sickness. The experimental environment consisted of one real environment and six VR environments. Participants wore an HMD and held controllers in both hands, freely navigating each space and initiating transitions at arbitrary timings. The user study was conducted in a seated position, with locomotion in VR performed via joystick-based input, allowing the study to focus on how passing through a transition space is perceived as movement rather than on physical walking. Three conditions were prepared: a Mirror Condition using MirroringWay, a Board Condition in which a board-shaped object served as the transition medium, and a Baseline Condition in which spaces were switched using only a fade effect. In both the Mirror and Board Conditions, participants transitioned by passing through a corridor-like space. After each condition, participants completed brief subjective evaluations, followed by comparative questions after all conditions. Screen recordings were also collected to observe participant behavior. These evaluations served as supplementary data that indicate the acceptability of the proposed method as a cross-space transition experience.

Through the demo and user study, several tendencies were observed regarding how the mirror-based transition experience was received. In subjective evaluations, the Mirror Condition showed significantly higher ratings for enjoyment compared to the other

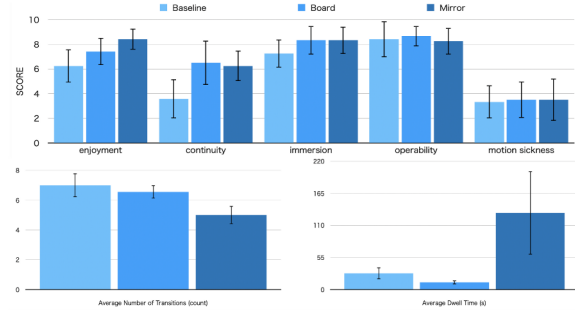


Figure 4: Subjective ratings and behavioral measures for each transition condition. Error bars indicate 95% confidence intervals.

conditions. Regarding perceived continuity, both the Board Condition and the Mirror Condition were rated significantly higher than the Baseline Condition. No statistically significant differences were found among conditions for immersion, operability, or motion sickness. In free-form responses, multiple participants commented that “there was a sense of narrative in the mirror imagery turning into a passage,” suggesting that the mirror-mediated transition was intuitively understandable. Others noted that “seeing myself in the mirror made me feel like I was in that world,” highlighting the simultaneous perception of one's self-image and the surrounding environment. Analysis of screen-recorded data further indicated that, in the Mirror Condition, participants tended to transition less frequently and remain longer within each space. These findings are positioned as qualitative evidence that mirror-mediated transitions can function as a viable transition experience, rather than as a claim of superiority over other methods.

5 DISCUSSION

This paper proposed MirroringWay, a mirror-mediated transition method, and presented a demo targeting experiences involving transitions across multiple spaces. This section discusses how mirror-based transitions function within such experiences.

Observations suggest that the depth impression produced by facing mirrors and the transformation of mirror imagery into a passable corridor make the emergence of a transition space perceptually convincing. Perceiving one's self-image together with the surrounding environment through the mirror also appears to enhance the appeal of the transition experience. Behavioral tendencies observed in screen-recorded data further suggest that MirroringWay influenced not only subjective impressions but also participants' behavior within the space. MirroringWay is positioned not merely as a technique for generating a transition corridor, but as a perceptual trigger for world transitions. Passing through a mirror evokes a familiar metaphor of “crossing to the other side,” which functions effectively as an entry point for transition experiences. As illustrated in Figure 2, MirroringWay is designed as a transition trigger capable of connecting multiple worlds beyond a fixed pair of spaces. Future work includes diversifying mirror and corridor representations to further support intuitive inter-world navigation.

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